

PRATHAM MALLICK

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Summary

B.Tech student in Electrical and Electronics Engineering with 6 months of industry experience in application automation and software quality engineering at Siemens EDA. Experienced in developing automated test solutions using Java, Selenium, Playwright, Cucumber.js, and Jenkins, with hands-on exposure to application maintenance, regression testing, CI/CD, and Agile development. Strong analytical, debugging, and problem-solving skills with a passion for automation engineering.

Education

B.Tech in Electrical and Electronics Engineering 2022 – 2026
Vellore Institute of Technology, Vellore CGPA: 8.87/10.0

Work Experience

Software Intern Jan – Jul 2026
Siemens EDA

- Developed and maintained 60+ automated test scripts for Capital Systems Architect, Capital Systems Modeler, and Capital Workspace using Java, JavaScript, Selenium, Playwright, Puppeteer, and Cucumber.js.
- Designed end-to-end UI automation for Capital Workspace features within the MARS Automation Framework, building reusable utilities and improving automation reliability through debugging and test stabilization.
- Authored 50+ CSV-driven automation workflows and custom Java automation functions to validate complex engineering design workflows across desktop applications.
- Executed manual, regression, and exploratory testing across 3 Capital releases, identifying and documenting 10+ defects that were validated and resolved prior to release.
- Integrated automated test suites into Jenkins CI/CD pipelines and supported automation execution, pipeline monitoring, and release validation activities.
- Collaborated with QA engineers, developers, product owners, and engineering managers during test bashes and release validation to ensure product quality and system stability.
- Worked in Agile/Scrum environments, collaborating with cross-functional teams and utilizing Jira and Atlassian tools for defect logging, tracking, sprint management, and release coordination.
- Worked with Git, PostgreSQL, Oracle databases, and migration workflows while supporting automation, quality assurance, and performance validation efforts.

Research Intern May – Jul 2025
NCPRE, IIT Bombay

- Gained hands-on experience with sputtering processes, thin-film deposition, and characterization of Indium Tin Oxide (ITO).
- Worked with high-vacuum systems and studied the electrical and optical properties of deposited thin films.

VLSI Design Intern May – Jun 2025
Maven Silicon

- Designed and implemented an SPI RTL module while learning digital design and verification fundamentals.

Skills

Languages: Java, Python, JavaScript, SQL, React.js, Embedded C, MATLAB, Verilog HDL
Testing: Selenium, Playwright, Puppeteer, Cucumber.js, Manual Testing, Regression Testing, Functional Testing, Debugging
Tools: Git, Jenkins, MySQL, PostgreSQL, Oracle Database, AWS
Methodologies: Agile/Scrum, CI/CD, Defect Tracking, Incident Resolution, Root Cause Analysis, Software Quality Assurance
Engineering: MATLAB Simulink, Keil MDK, Arduino IDE, OrCAD, PSpice
AI/ML: Predictive Analytics, NLP, LLMs, RAG, Prompt Engineering, TensorFlow, PyTorch, Scikit-learn
Soft Skills: Creativity, Problem solving, Communication, Leadership, Troubleshooting

Projects

Health Monitoring of EV Motors using Soft Computing Academic Project

- Developed data-driven models to assess motor health and predict failures in Electric Vehicles.
- Aims to enhance reliability, reduce downtime, and enable predictive maintenance in EV systems.

DC Motor Control System for Microgrid Application Power Systems Simulation Project

- Simulated a closed-loop DC motor system in MATLAB Simulink to analyze performance under dynamic load conditions.
- Designed and tuned a PI controller to regulate motor speed and improve system stability.
- Validated system performance using state-space and transfer function modeling techniques.

Certifications and Achievements

Certifications: Google AI Professional Certificate (Coursera), AI/ML for Geodata Analysis (ISRO), Full Stack Web Development (Udemy)
Hackathons: 3rd place in Wired-In Hackathon, participated in HackBattle - Gravitas 2023, and Bolt 2.0 2024.
Challenges: Participated in TATA Imagination Challenge 2024 and NCIIPC-AICTE Pentathlon 2025.